

Technology Stack

Date	1 July 2026
Team ID	—
Project Name	Rising Waters – Flood Prediction System
Maximum Marks	2 Marks

The table below details the specific technologies chosen for each component of the Rising Waters architecture.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5 + Jinja2 templates, custom CSS (static/css/style.css)	Simple, dependency-free UI that renders quickly and is easy for a small team to maintain
2	Backend / Server-Side	Python 3, Flask	Lightweight web framework, integrates directly with the Python ML stack (scikit-learn, joblib)
3	Machine Learning	scikit-learn (Decision Tree, Random Forest, KNN), XGBoost	Well-established classifiers; multiple models compared and the best one selected automatically
4	Data Processing	pandas, numpy	Efficient tabular data handling for preprocessing, scaling and feature engineering
5	Visualization / EDA	matplotlib, seaborn	Used to generate distribution, boxplot, heatmap, pairplot and evaluation charts
6	Model Persistence	joblib (flood_model.pkl, scaler.pkl), JSON (metadata.json)	Fast, standard serialization for scikit-learn/XGBoost models so Flask can load them at startup
7	Database / Data Storage	Flat file — data/flood_data.csv	Dataset size and project scope don't require a full DBMS; keeps the pipeline simple and portable
8	Deployment (optional)	IBM Cloud Code Engine / gunicorn	Allows the trained model to be exposed as a public URL for disaster-management teams
9	Version Control	Git	Tracks changes to code, notebooks and templates across the team